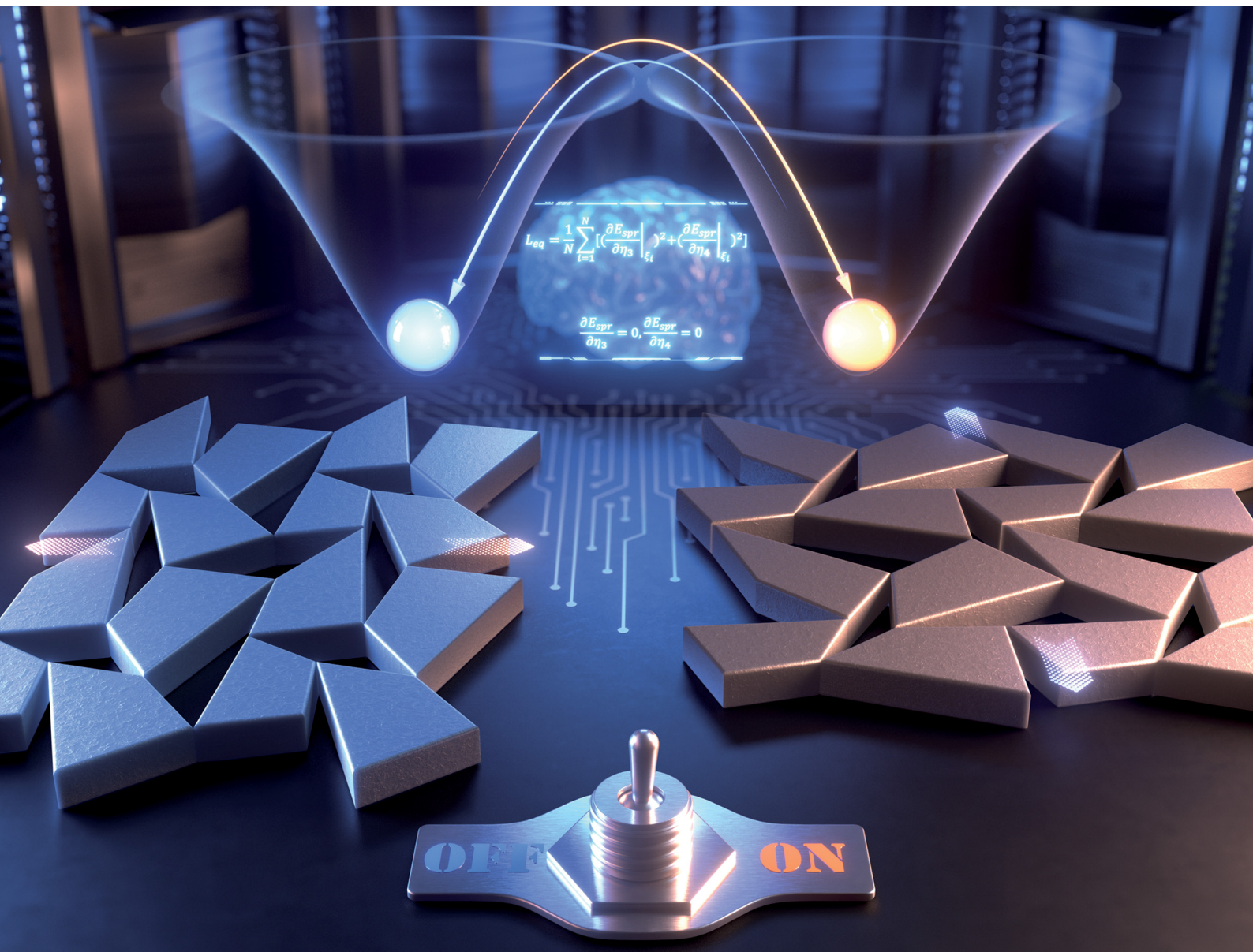


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Data-free physics-informed inverse programming of bistable kirigami energy landscapes

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Kirigami metamaterials, formed by introducing cuts into planar sheets, can exhibit bistability through geometry-dependent energy landscapes. Designing structures with prescribed energy barriers and target deformation states is therefore essential for programmable mechanical functionality. Here, we present a physics-informed neural network (PINN) framework that does not require pre-collected labeled training datasets and unifies forward prediction and inverse programming of bistable kirigami energy landscapes by embedding equilibrium conditions, energy formulations, and geometric compatibility directly into the learning objective. In the forward setting, the framework predicts continuous energy landscapes with coefficients of determination above 0.99 and barrier errors below 0.1%. In the inverse setting, it identifies kirigami geometries from fully prescribed energy curves with barrier errors below 5%, and further extends to a minimally specified setting in which only the target energy barrier and zero-energy stable states are given. In this underdetermined case, the framework autonomously infers the full energy landscape while achieving a mean barrier error of 0.4%. Finite element analysis and experiments on 3D-printed prototypes confirm that the programmed ordering of energy barriers is preserved. By assembling unit cells with different programmed barriers, we further demonstrate sequential actuation, in which units with lower barriers transform first during tensile loading. This work establishes an efficient physics-informed approach for programming bistable energy landscapes in planar kirigami systems, with potential applications in deployable structures, soft robotic actuators, and impact mitigation devices.

New concepts

We introduce a new design paradigm for bistable mechanical metamaterials by recasting design from geometry specification to direct energy landscape programming. Conventional kirigami design approaches prescribe geometry and evaluate the resulting response, making inverse design indirect and inefficient. Here, we instead treat the energy landscape itself as the primary design object, directly specifying functional targets such as stable states and energy barriers. By embedding mechanical equilibrium, energy formulations, and geometric compatibility into a physics-informed neural network, the framework enables data-free inverse programming that maps desired energy characteristics to feasible kirigami geometries. This shift transforms bistability from an emergent consequence of geometry into a directly programmable property defined in the energy domain, enabling multiple feasible designs and controlled functionalities such as sequential actuation. More broadly, this energy-first paradigm establishes a general route for programming mechanical behavior across diverse bistable metamaterial systems.

1. Introduction

Mechanical metamaterials derive their functionality not from material composition but from structural geometry, enabling unconventional mechanical responses such as negative Poisson's ratio, large reversible deformations, and multistability.^{1–6} Among the various metamaterial platforms, kirigami architectures created by introducing strategically placed cuts into planar sheets have attracted particular attention due to their scalability, geometric richness, and ability to undergo large, nonlinear deformations while remaining lightweight and manufacturable.^{7–12} A notable feature accessible through kirigami geometry is bistability, which allows mechanical systems to store, release, and reconfigure elastic energy in a controlled manner between two distinct stable configurations,^{13–22} forming the basis for deployable structures, adaptive systems, and sequential mechanical operations.^{23–29}

To design kirigami-based mechanical systems with prescribed functionality, inverse design problems are often formulated as constrained optimization tasks.^{30–32} In these approaches,

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geometric compatibility, non-overlap conditions, and boundary constraints are explicitly imposed to ensure physically realizable configurations.^{7,33–38} Although such methods enable rational design of bistable and shape-morphing kirigami structures, their reliance on hard constraints and local search strategies can limit exploration of the design space, making the solutions sensitive to initialization and restricting access to alternative feasible designs.

To address these limitations, data-driven and machine-learning-based approaches have recently been introduced.^{39,40} By training surrogate models or neural networks on pre-computed datasets, these methods approximate the mapping between structural parameters and target responses, reducing the need for repeated optimization.^{32,41–47} However, their effectiveness depends on the availability of large, well-curated datasets that are costly to obtain through extensive simulations or experiments. Moreover, data-driven models often show limited generalization beyond the sampled design space and lack explicit mechanisms to enforce the strict physical and geometric constraints inherent to kirigami systems.

In bistable kirigami systems, functional behavior is determined not only by the presence of multiple equilibrium states but also by how elastic energy is distributed along the deformation pathway connecting them (Fig. 1A).³⁰ Consequently, stability and transition characteristics are governed by the underlying energy landscape, particularly by the relative depths of energy

minima and the barriers separating them. However, in conventional approaches, energy information is typically obtained only at discrete deformation states encountered during simulations or optimization. Likewise, data-driven approaches infer target responses from finite datasets without explicitly representing the continuous energy profile that governs deformation. As a result, key inverse design objectives for bistable systems, such as prescribed stable states or energy barriers, remain difficult to impose directly in a systematic and physically transparent manner.

Here, we propose a physics-informed inverse programming framework for designing bistable kirigami energy landscapes without requiring pre-collected labeled training datasets (Fig. 1B). Instead of relying on precomputed datasets or strictly enforced optimization rules, the proposed approach incorporates equilibrium conditions, energy formulations, and geometric compatibility into a physics-informed neural network (PINN) through physically motivated loss functions.^{48–53} This soft-constraint formulation guides the design process using physical principles while enabling flexible exploration of the design space. By learning continuous energy landscapes along the deformation coordinate from physics-based residuals and design specifications, the framework enables inverse programming from minimal specifications, such as target stable states or prescribed energy barriers, while forward prediction and inverse programming are addressed within a unified

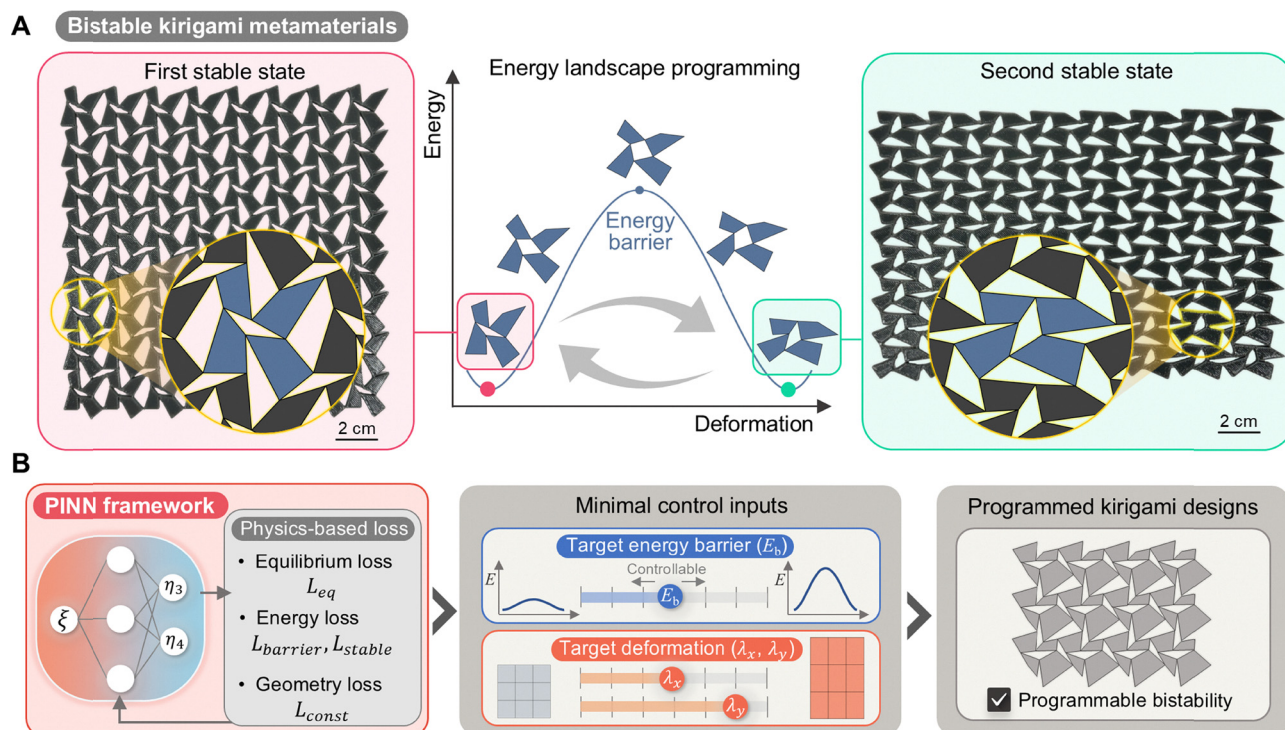


Fig. 1 Bistable kirigami metamaterials and physics-informed prediction and inverse programming framework. (A) Schematic of a bistable kirigami metamaterial designed to exhibit two geometrically distinct stable states, and the corresponding double-well energy landscape along the deformation coordinate. (B) Schematic of the PINN-based framework for forward prediction and inverse programming, in which minimal control inputs, namely the target energy barrier (E_b) and target deformation (λ_x, λ_y), are translated into programmed kirigami designs through physics-informed loss functions (equilibrium, energy, and geometric losses).

architecture. Using this framework on the rotating-squares kirigami architecture,^{12,30} we demonstrate forward prediction of energy landscapes, inverse design from fully prescribed energy curves, and inverse programming from target energy barriers. The inverse-programmed structures are validated through finite element analysis and experiments on 3D-printed prototypes, both confirming that the monotonic ordering of programmed barriers is preserved. By further assembling unit cells with identical macroscopic deformation but distinct programmed barriers, we demonstrate sequential actuation in which units with lower barriers transform first under uniform loading. This work establishes an efficient physics-informed approach for programming bistable energy landscapes in planar kirigami systems, with potential applications in deployable structures,^{25,26} soft robotics,^{54,55} and impact-mitigation devices.

2. Methodology

2.1. Bistable kirigami system and energy landscape formulation

We consider a class of bistable planar kirigami architectures based on a perturbed rotating-squares mechanism,^{12,56,57} following the formulation established by Peng *et al.*³⁰ A unit cell consists of four rigid quadrilateral panels (indexed by $i = 1, \dots, 4$) separated by four quadrilateral slits, whose geometry is described by sixteen in-plane boundary vectors $\{s_i, t_i, u_i, v_i\}_{i=1}^4$, where each vector represents an edge segment defining the boundaries of the panels and slits within the unit cell (Fig. 2A). For the unit cell to form a compatible bistable pattern, these vectors must satisfy a set of equality constraints including slit-closure, panel-closure, and periodicity conditions in both the reference and deformed states. In the deformed state, each panel i undergoes a rigid in-plane rotation $R(\phi_i)$, where ϕ_i denotes the rotation angle of the i -th panel and $R(\cdot)$ denotes the planar rotation matrix. The unit cell tiles the plane *via* Bravais lattice vectors $(\mathbf{r}_1^R, \mathbf{r}_2^R)$ and $(\mathbf{r}_1^D, \mathbf{r}_2^D)$, where $\mathbf{r}_1^R$ and $\mathbf{r}_2^R$ correspond to the

reference configuration and $\mathbf{r}_1^D$ and $\mathbf{r}_2^D$ correspond to the deformed configuration, which define the periodic tiling of the unit cell in each configuration. The complete set of equality and inequality constraints, together with the design formula that parameterizes all dependent vectors as functions of the free variables, is provided in SI, Section S1.

By solving the equality constraints, all sixteen boundary vectors are expressed as functions of four scalar rotation angles $(\phi_1, \phi_2, \phi_3, \phi_4)$, two free boundary vectors $(\mathbf{s}_1, \mathbf{s}_2)$, and the prescribed Bravais lattice vectors. For a given set of lattice vectors, the remaining free design parameters are collected into an eight-component design vector, defined as follows:

$$\mathbf{x} = (s_{11}, s_{12}, s_{21}, s_{22}, \phi_1, \phi_2, \phi_3, \phi_4) \in \mathbb{R}^8 \quad (1)$$

Here, (s_{11}, s_{12}) and (s_{21}, s_{22}) denote the Cartesian components of the independent boundary vectors $\mathbf{s}_1$ and $\mathbf{s}_2$, respectively. Macroscopic deformation between the two stable states is quantified by the lattice stretch ratios, defined as:

$$\lambda_x = \frac{\|\mathbf{r}_1^D\|}{\|\mathbf{r}_1^R\|}, \lambda_y = \frac{\|\mathbf{r}_2^D\|}{\|\mathbf{r}_2^R\|} \quad (2)$$

which represent the directional stretches along the two lattice vectors between the reference and deformed configurations.

Geometric compatibility guarantees the existence of two equilibrium states but does not by itself ensure bistability, which additionally requires an energy barrier separating the two states along the deformation pathway. To characterize this behavior, we introduce a reduced-order elastic energy model^{30,58} in which the corner points of the kirigami pattern are connected by linear springs of zero rest length and unit stiffness (Fig. 2B).

In the reference configuration, the slit boundary vectors form closed loops such that the sum of the vectors associated with each slit vanishes. Under deformation, these loop-closure conditions are generally violated due to the mismatch induced by panel rotations and periodic constraints, leading to geometric

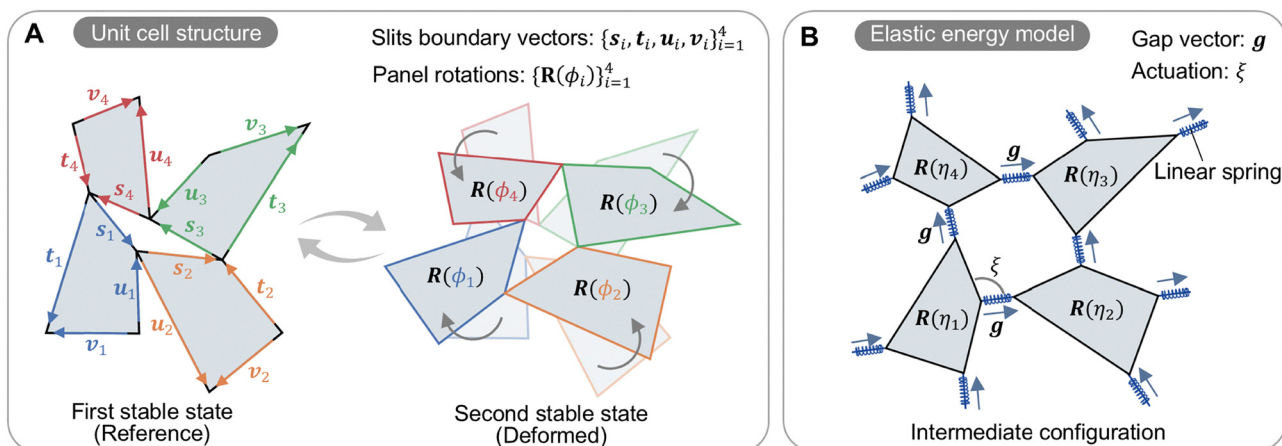


Fig. 2 Unit cell geometry and energy formulation of the bistable kirigami system. (A) Unit cell geometry in the first stable (reference) and second stable (deformed) states, showing the boundary vectors $(\mathbf{u}, \mathbf{v}, \mathbf{s}, \mathbf{t})$ and panel rotations $R(\phi_1) - R(\phi_4)$. (B) Intermediate configuration during actuation, illustrating the actuation coordinate ξ , panel rotations $R(\eta_1) - R(\eta_4)$ and gap vectors $\mathbf{g}$ generated by geometric incompatibility between adjacent panels, which define the elastic energy of the system.

incompatibility. During actuation, this incompatibility between panels produces gap vectors $\mathbf{g}_{ij}(\eta)$ across slits, where $\mathbf{g}_{ij}(\eta)$ denotes the incompatibility associated with adjacent panels i and j under the panel rotation state $\eta = (\eta_1, \eta_2, \eta_3, \eta_4)$, measured as the mismatch resulting from the violation of loop-closure conditions induced by their relative rotations and lattice constraints. The elastic energy is then given by

$$E_{\text{spr}}(\eta) = \sum_{ij} \|\mathbf{g}_{ij}(\eta)\|^2 \quad (3)$$

where the summation extends over all adjacent panel pairs. After minimizing over translations and lattice vectors, the energy depends only on the panel rotation variables (SI, Section S1).

The actuation is parameterized by a scalar coordinate ξ by fixing the first panel rotation and defining

$$\eta_1 = \phi_1, \eta_2 = \phi_1 + \xi, \xi \in [0, \phi_2 - \phi_1] \quad (4)$$

where $\eta = (\eta_1, \eta_2, \eta_3, \eta_4)$, and η_3 and η_4 are internal rotation variables that adapt to minimize the elastic energy. The actuation energy landscape is defined as:

$$E_{\text{act}}(\xi) = \min_{\eta_3, \eta_4} E_{\text{spr}}(\phi_1, \phi_1 + \xi, \eta_3, \eta_4) \quad (5)$$

By construction, the two stable states satisfy $E_{\text{act}}(0) = E_{\text{act}}(\phi_2 - \phi_1) = 0$, since the slit boundary vectors form closed loops in both configurations, resulting in zero geometric incompatibility. For intermediate values of ξ , these loop-closure conditions are generally violated, leading to nonzero incompatibility and thus

$E_{\text{act}}(\xi) > 0$, which gives rise to a characteristic double-well energy landscape. The bistable energy barrier is defined as:

$$E_b = \max_{\xi \in (0, \phi_2 - \phi_1)} E_{\text{act}}(\xi) \quad (6)$$

This energy formulation provides the foundation for both forward prediction and inverse design of bistable kirigami systems, as developed in the following section.

2.2. PINN framework for bistable kirigami

2.2.1. Forward prediction. The key computational task in evaluating $E_{\text{act}}(\xi)$ (eqn (5)) is determining the internal rotation angles η_3 and η_4 that minimize the elastic energy at each actuation step. This corresponds to solving a sequence of inner energy minimization problems along the actuation coordinate. Rather than solving this optimization repeatedly using conventional numerical methods, we employ a PINN⁴⁸ to learn the equilibrium rotation paths $\eta_3(\xi)$ and $\eta_4(\xi)$ as continuous functions of the actuation coordinate.

The neural network takes the scalar actuation coordinate $\xi \in [0, \phi_2 - \phi_1]$ as input and outputs the internal rotation angles (η_3, η_4) . The network consists of fully connected hidden layers with hyperbolic tangent activation functions, and the outputs are scaled to enforce physically admissible rotation ranges $\eta_3, \eta_4 \in [-\pi, \pi]$ (Fig. 3). The detailed network architecture, hyperparameters, and architecture ablation analysis are provided in SI, Section S2.

Training is performed without labeled target data by enforcing the first-order necessary conditions for energy minimization with respect to the internal rotations, where the corresponding

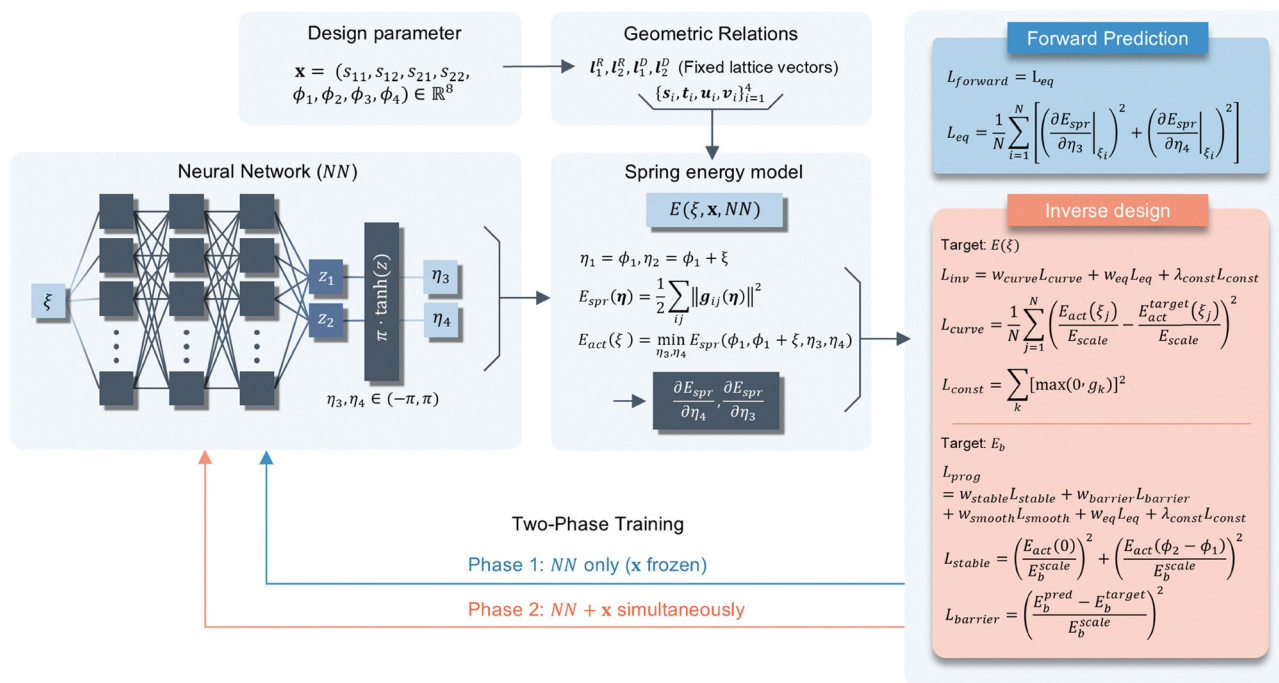


Fig. 3 PINN framework architecture, loss formulations, and training procedure. Schematic of the PINN framework showing the neural network architecture (input: ξ ; outputs: η_3, η_4 ; design parameters: $\mathbf{x} \in \mathbb{R}^8$), the spring energy model $E(\xi, \mathbf{x}, \text{NN})$, loss functions for forward prediction (L_{forward}) and inverse design ($L_{\text{inv}}, L_{\text{prog}}$), and the two-phase training procedure.

equilibrium residuals are driven toward zero:

$$\frac{\partial E_{\text{spr}}}{\partial \eta_3} = 0, \frac{\partial E_{\text{spr}}}{\partial \eta_4} = 0 \quad (7)$$

The training loss consists solely of the equilibrium residual, defined as $L_{\text{forward}} = L_{\text{eq}}$, where

$$L_{\text{eq}} = \frac{1}{N} \sum_{i=1}^N \left[\left(\frac{\partial E_{\text{spr}}}{\partial \eta_3} \Big|_{\xi_i} \right)^2 + \left(\frac{\partial E_{\text{spr}}}{\partial \eta_4} \Big|_{\xi_i} \right)^2 \right] \quad (8)$$

Here, N denotes the number of collocation points uniformly distributed over the actuation interval. The gradients are computed *via* automatic differentiation through the spring energy (eqn (3)) and the network outputs. Minimizing L_{eq} enforces the stationarity conditions (eqn (7)) at all collocation points, thereby enabling the network to learn the equilibrium deformation path without relying on any precomputed energy data.

2.2.2. Inverse design from full energy curves. We next consider the inverse problem of identifying a design vector $\mathbf{x}$ that reproduces a prescribed target actuation energy landscape $E_{\text{act}}^{\text{target}}(\xi)$. Unlike the forward problem, where the geometry is fixed, and only the internal rotation paths are learned, the inverse design task treats both the design vector $\mathbf{x}$ and the neural network parameters as trainable variables that are optimized simultaneously.

The neural network architecture is identical to that used in the forward prediction (Section 2.2.1), taking ξ as input and outputting (η_3, η_4) (Fig. 3). In addition, the eight components of the design vector $\mathbf{x} \in \mathbb{R}^8$ (eqn (1)) are treated as differentiable parameters and updated together with the network weights. Because $\mathbf{x}$ determines the corner vectors $\{\mathbf{s}_i, \mathbf{t}_i, \mathbf{u}_i, \mathbf{v}_i\}$ through the geometric relations of Section 2.1, the spring energy E_{spr} is differentiable with respect to both the network outputs and the design parameters. This enables end-to-end optimization of the mapping from geometry to the resulting energy landscape.

The total loss function for inverse design consists of three components:

$$L_{\text{inv}} = w_{\text{curve}} L_{\text{curve}} + w_{\text{eq}} L_{\text{eq}} + \lambda_{\text{const}} L_{\text{const}} \quad (9)$$

The curve-matching loss enforces agreement between the predicted and target energy landscapes, defined as:

$$L_{\text{curve}} = \frac{1}{N} \sum_{j=1}^N \left(\frac{E_{\text{act}}(\xi_j)}{E_{\text{scale}}} - \frac{E_{\text{act}}^{\text{target}}(\xi_j)}{E_{\text{scale}}} \right)^2 \quad (10)$$

where $E_{\text{scale}} = \max_j E_{\text{act}}^{\text{target}}(\xi_j)$ is a normalization constant. The equilibrium loss L_{eq} is identical to that of the forward problem (eqn (8)) and enforces the stationarity conditions $\partial E_{\text{spr}} / \partial \eta_3 = 0$, $\partial E_{\text{spr}} / \partial \eta_4 = 0$. The constraint loss penalizes violations of the geometric inequality constraints introduced in Section 2.1, defined as:

$$L_{\text{const}} = \sum_k [\max(0, -g_k)]^2 \quad (11)$$

where $g_k \geq 0$ represents the k -th inequality constraint ensuring convexity and non-self-intersection.

Training proceeds in two phases. In Phase 1, the design parameters are frozen and only the network weights are updated using L_{eq} , allowing the network to learn an approximate equilibrium deformation path for the initial geometry. In Phase 2, both the network weights and the design parameters are updated simultaneously, the network is trained using L_{eq} , while the design parameters are optimized using the full loss L_{inv} (eqn (9)). This staged training prevents unstable design updates before a meaningful deformation field is established.

Because the mapping from geometry to energy landscape is generally many-to-one, the inverse problem may admit multiple solutions. To reduce sensitivity to initialization, we employ a multi-start strategy: one initial design is taken from the reference configuration reported previously,³⁰ while three additional starting points are generated by randomly sampling geometrically valid structures. The final design is selected based on the best curve-matching accuracy and constraint satisfaction. The detailed training hyperparameters are provided in SI, Section S2.

This inverse design framework provides a unified physics-informed inverse-design approach that simultaneously learns the equilibrium deformation paths and identifies structural parameters consistent with a prescribed energy landscape.

2.2.3. Inverse programming from partial energy specifications. The full-curve inverse design framework in Section 2.2.2 requires a complete target energy landscape as input. We now extend this formulation to an underdetermined design setting in which only partial energy specifications are provided. Specifically, the design input consists of a target energy barrier $E_{\text{b}}^{\text{target}}$ together with the requirement that the energy vanishes at the two stable states, $E_{\text{act}}(0) = E_{\text{act}}(\phi_2 - \phi_1) = 0$. The detailed shape of the energy landscape between these states, including the location of the barrier, is not prescribed but instead inferred by the model.

The network architecture and the treatment of the design vector $\mathbf{x}$ as differentiable parameters remain identical to those in Section 2.2.2 (Fig. 3). The key difference lies in the loss function, which replaces the dense curve-matching loss L_{curve} (eqn (10)) with pointwise energy conditions at the two stable states and the barrier. The total loss function is defined as:

$$L_{\text{prog}} = w_{\text{stable}} L_{\text{stable}} + w_{\text{barrier}} L_{\text{barrier}} + w_{\text{eq}} L_{\text{eq}} + w_{\text{smooth}} L_{\text{smooth}} + \lambda_{\text{const}} L_{\text{const}} \quad (12)$$

The stable-state loss enforces zero energy at the two end-points of the actuation interval, defined as:

$$L_{\text{stable}} = \left(\frac{E_{\text{act}}(0)}{E_{\text{b}}^{\text{scale}}} \right)^2 + \left(\frac{E_{\text{act}}(\phi_2 - \phi_1)}{E_{\text{b}}^{\text{scale}}} \right)^2 \quad (13)$$

The barrier loss drives the predicted energy barrier toward the target value, defined as:

$$L_{\text{barrier}} = \left(\frac{E_{\text{b}}^{\text{pred}} - E_{\text{b}}^{\text{target}}}{E_{\text{b}}^{\text{scale}}} \right)^2 \quad (14)$$

where $E_b^{\text{pred}} = \max_{\xi} E_{\text{act}}(\xi) - \min(E_{\text{act}}(0), E_{\text{act}}(\phi_2 - \phi_1))$ and E_b^{scale} is a normalization constant. Including L_{stable} together with L_{barrier} is essential. In the full-curve formulation (Section 2.2.2), the dense curve-matching loss L_{curve} implicitly enforces near-zero energy at the endpoints. In contrast, the present formulation contains no dense energy supervision, and the equilibrium loss L_{eq} alone only enforces stationarity. Because the barrier is defined as a relative quantity, the target barrier could otherwise be satisfied even if the entire energy landscape were shifted. The stable-state loss anchors the energy scale and prevents this degeneracy.

The equilibrium loss L_{eq} (eqn (8)) and the constraint loss L_{const} (eqn (11)) are retained from the previous formulations. Because only the endpoints and barrier height are specified, the deformation path between the two stable states remains largely unconstrained. To ensure physically plausible continuous deformation paths, we introduce a smoothness regularization term, defined as:

$$L_{\text{smooth}} = \frac{1}{N} \sum_{j=1}^{N-1} \left[(\eta_3(\xi_{j+1}) - \eta_3(\xi_j))^2 + (\eta_4(\xi_{j+1}) - \eta_4(\xi_j))^2 \right] \quad (15)$$

which penalizes abrupt variations in the predicted rotation paths. This term is unnecessary in the full-curve formulation, where the dense target energy curve already constrains the intermediate deformation path.

Training follows the same two-phase procedure described in Section 2.2.2. In Phase 1, only the network weights are updated using $L_{\text{eq}} + L_{\text{smooth}}$. In Phase 2, both the network parameters and the design parameters are optimized using the full loss L_{prog} (eqn (12)). The multi-start initialization strategy is also identical. Detailed training hyperparameters are provided in SI, Section S2.

Together, Sections 2.2.1–2.2.3 form a unified PINN framework for bistable kirigami energy landscapes. The framework progresses from forward prediction for a fixed geometry, to inverse design from a fully specified energy curve, and finally to inverse programming from minimal energy specifications. All three formulations share the same network architecture and physics-informed learning strategy, differing only in the loss formulation and the set of optimized variables.

2.3. Finite element analysis (FEA)

To validate the PINN-predicted energy landscapes, finite element analyses were performed using the ABAQUS/Implicit solver (Dassault Systèmes). In contrast to the reduced-order spring model introduced in Section 2.1, where panels are treated as rigid bodies connected by point hinges, the finite element model represents the kirigami structure as a two-dimensional deformable continuum with hyperelastic panels connected by compliant hinges. This higher-fidelity representation provides an independent benchmark for evaluating the PINN-based predictions and the extent to which the programmed barrier hierarchy is preserved in a continuum setting.

Because ideal point hinges cannot be directly implemented in a finite element mesh, each hinge was converted into a finite-length line hinge using a bisector-based construction method. This approach ensures a symmetric hinge geometry and uniform hinge length across the unit cell while avoiding artificial stress concentrations. The detailed hinge construction procedure is described in SI, Section S3.

The kirigami panels were modeled as isotropic hyperelastic solids using a Mooney–Rivlin constitutive law. The material parameters were calibrated from uniaxial tensile tests performed on dogbone specimens fabricated by three-dimensional (3D) printing using thermoplastic polyurethane filament (TPU-95A, eSUN). Details of the specimen fabrication, tensile testing, and parameter fitting procedure are provided in SI, Section S3.

Finite element simulations were performed under large-deformation conditions to capture the nonlinear folding behavior and snap-through instability associated with bistable kirigami structures. Actuation was applied through panel-level reference points using distributed coupling constraints, and prescribed rotations were used to drive the structure across the energy barrier. Local mesh refinement was introduced near the hinge regions to accurately resolve hinge bending and stress localization, while self-contact was defined along all panel edges to prevent interpenetration during large deformation. Detailed descriptions of the mesh construction, boundary conditions, and analysis settings are provided in SI, Section S3.

The finite element validation focuses on the five structures inverse-designed *via* the partial energy specification framework (Section 2.2.3). These structures share the same macroscopic deformation but are programmed with different energy barriers. The resulting strain-energy curves are compared with the PINN predictions to evaluate whether the predicted ordering of energy barriers is preserved in the continuum model.

2.4. Experimental methods

To experimentally validate the energy landscapes predicted by the PINN-based inverse programming framework, kirigami specimens were fabricated using fused deposition modeling (FDM) 3D printing with TPU filament (TPU-95A, eSUN). The specimens were designed based on the vertex coordinates of the deformed-state unit cells obtained from the inverse programming results.

Two types of specimens were prepared. For energy-barrier validation, the kirigami unit cell was arranged in a 2×2 periodic array. To demonstrate sequential deformation behavior, specimens consisting of a 4×2 array of unit cells were fabricated. All specimens were printed with 100% infill density using the same material and printing conditions.

Uniaxial tensile tests were conducted using a universal testing machine (UTA, YEONJIN S-Tech) under displacement control. Before testing, each specimen was aligned to the reference configuration, and tensile loading was applied to drive the structure toward the programmed deformed configuration. The actuation response and reaction forces were recorded to evaluate the programmed energy barrier and deformation behavior. Detailed descriptions of the specimen

geometry, hinge construction for 3D printing, and tensile testing setup are provided in SI Section S4.

3. Results and discussion

3.1. Forward prediction of energy landscapes

We first evaluate the forward prediction capability of the PINN framework by comparing predicted energy landscapes with ground-truth solutions obtained *via* direct numerical minimization at each actuation step. Two sets of test cases are considered to assess the model performance under varying energy barriers and macroscopic deformation conditions.

In the first set, the macroscopic deformation is fixed at $(\lambda_x, \lambda_y) = (1.2, 0.8)$ while the energy barrier varies across five levels, $E_b = 0.001, 0.002, 0.003, 0.004$, and 0.005 , each corresponding to a distinct kirigami geometry (Fig. 4A and B). In the second set, the energy barrier is fixed at $E_b = 0.003$ while the macroscopic deformation varies across six configurations spanning uniaxial, equibiaxial, and shear-like deformation states, $(\lambda_x, \lambda_y) = (0.8, 1.2), (1.0, 1.0), (1.0, 1.2), (1.2, 0.8), (1.2, 1.0)$ and $(1.2, 1.2)$ (Fig. 4C and D). After removing the overlapping case, a total of ten distinct kirigami structures are obtained. The corresponding design parameters are reported in SI, Section S5.

Fig. 4A shows the reference and deformed configurations of the five structures, illustrating how different energy barriers lead to distinct unit-cell geometries under the same macroscopic deformation. The predicted energy landscapes (solid lines) closely match the analytical ground truth (discrete points) across the entire actuation range (Fig. 4B), accurately reproducing both the double-well energy profiles and the magnitudes of the energy barriers.

Fig. 4C shows the reference and deformed configurations for the six deformation cases, highlighting the diversity of kirigami geometries associated with different macroscopic deformation states. The predicted energy landscapes (Fig. 4D) remain in excellent agreement with the analytical solutions despite substantial variation in the actuation range and energy landscape shapes.

Across all ten test cases, the predicted energy landscapes achieve a coefficient of determination $R^2 > 0.99$ and a relative error in the energy barrier below 0.1%, confirming the accuracy and robustness of the forward prediction framework. The convergence behavior of the equilibrium loss used for training is provided in SI, Section S2.

3.2. Inverse design from full energy curves

We next evaluate the inverse design capability of the PINN framework, in which the network identifies a kirigami geometry that reproduces a prescribed target energy landscape. The same ten test cases used in Section 3.1 are considered here, with the ground-truth energy curves serving as target inputs to the inverse formulation described in Section 2.2.2.

To verify the inverse-designed structures, two independent assessments are performed. First, the energy curve predicted by the PINN for the obtained design vector $\mathbf{x}$ is compared with the

target curve. Second, the energy landscape corresponding to the obtained $\mathbf{x}$ is recomputed *via* direct numerical minimization, providing a verification independent of the PINN approximation. Both assessments yield nearly identical results, confirming the self-consistency of the framework (SI, Section S5). The PINN-predicted curves are therefore used as the primary metric in the following comparisons.

Fig. 4E and F show the inverse design results for the first set. The reference and deformed configurations of the five inverse-designed structures are shown in Fig. 4E. The predicted energy landscapes (solid lines) closely match the prescribed target curves (discrete points) across the entire actuation range (Fig. 4F).

Fig. 4G and H present the results for the second set. The corresponding reference and deformed configurations are shown in Fig. 4G. The predicted energy landscapes (Fig. 4H) accurately reproduce the target curves despite substantial variation in the actuation range and energy landscape shapes.

Across all ten cases, the inverse-designed energy landscapes achieve a coefficient of determination $R^2 > 0.99$ and a relative energy barrier error below 5% (SI, Section S5), confirming that the inverse framework reliably identifies geometries whose energy landscapes reproduce the prescribed targets. The convergence behavior of the curve-matching and equilibrium losses is provided in SI, Section S2. It is important to note that the inverse design formulation does not admit arbitrary target energy landscapes. The feasible energy landscapes are constrained by the geometric compatibility and mechanical equilibrium conditions encoded in the forward model, and the inverse framework performs reliably when the target curves lie within this physically realizable regime. In contrast, prescribing unphysical or strongly extrapolated energy curves may lead to infeasible designs or degraded accuracy.

Although the inverse formulation constrains only the energy landscape and not the geometry, the resulting structures exhibit strong geometric similarity to the original designs used in Section 3.1. When differences appear, they correspond to reflection-symmetric variants of the original geometry, which naturally arise from the one-to-many nature of the inverse problem. The obtained design parameters for all cases are reported in SI, Section S5.

3.3. Inverse programming of partial energy specifications

We next evaluate the inverse programming formulation under a more underdetermined setting that better reflects practical design scenarios, where only partial energy specifications are provided. Instead of prescribing the full energy curve as in Section 3.2, the formulation described in Section 2.2.3 receives only the target energy barrier E_b together with the constraint that the energy vanishes at the two stable states. The same ten test cases used in Sections 3.1 and 3.2 are considered to enable direct comparison.

Fig. 5A–E show the inverse programming results for the first set. The reference and deformed configurations of the five inverse-programmed structures are shown in Fig. 5A.

Forward prediction of energy landscapes

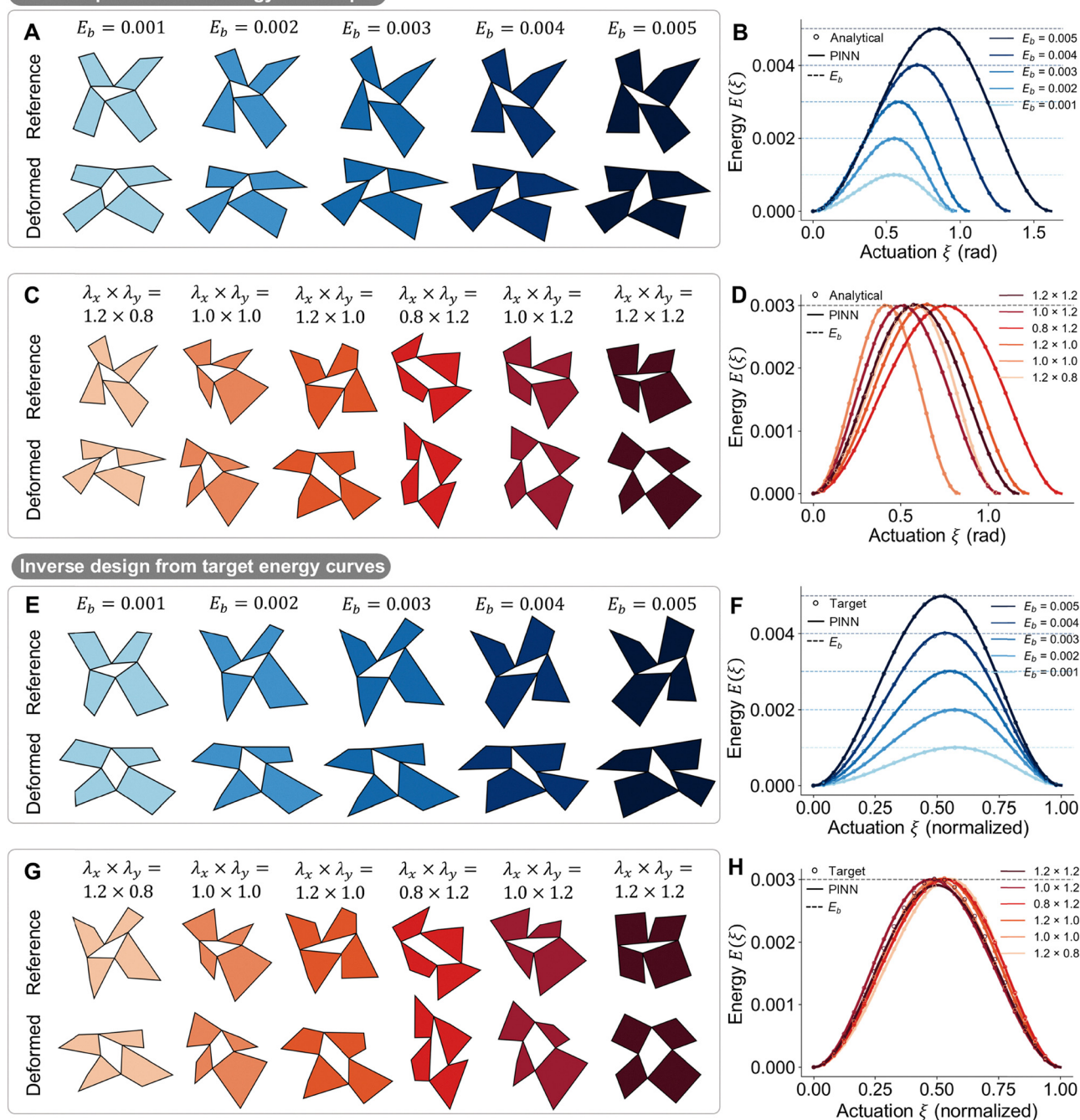


Fig. 4 Forward prediction and full-curve inverse design of bistable kirigami energy landscapes. (A)–(D) Forward prediction results. (A) Reference and deformed configurations of five kirigami structures with fixed macroscopic deformation and varying energy barriers. (B) Predicted energy landscapes (solid lines) compared with analytical ground truth (discrete points). (C) Reference and deformed configurations of six kirigami structures with fixed energy barrier and varying macroscopic deformations. (D) Predicted energy landscapes (solid lines) compared with analytical ground truth (discrete points). (E)–(H) Full-curve inverse design results. (E) Reference and deformed configurations of five structures obtained via full-curve inverse design under fixed macroscopic deformation with varying energy barriers. (F) Inverse-designed energy landscapes (solid lines) compared with prescribed target energy curves (discrete points). (G) Reference and deformed configurations of six structures under fixed energy barrier with varying macroscopic deformations. (H) Inverse-designed energy landscapes (solid lines) compared with prescribed target energy curves (discrete points).

The programmed energy landscapes (Fig. 5B) exhibit well-defined peaks that align closely with the prescribed barrier values, confirming accurate realization of the target energy barriers. The corresponding loss histories (Fig. 5C–E) show

consistent convergence of the equilibrium loss L_{eq} , stable-state loss L_{stable} , and barrier loss $L_{barrier}$ during training.

Fig. 5F–J present the results for the second set. The corresponding reference and deformed configurations are shown

Inverse programming from target energy barriers

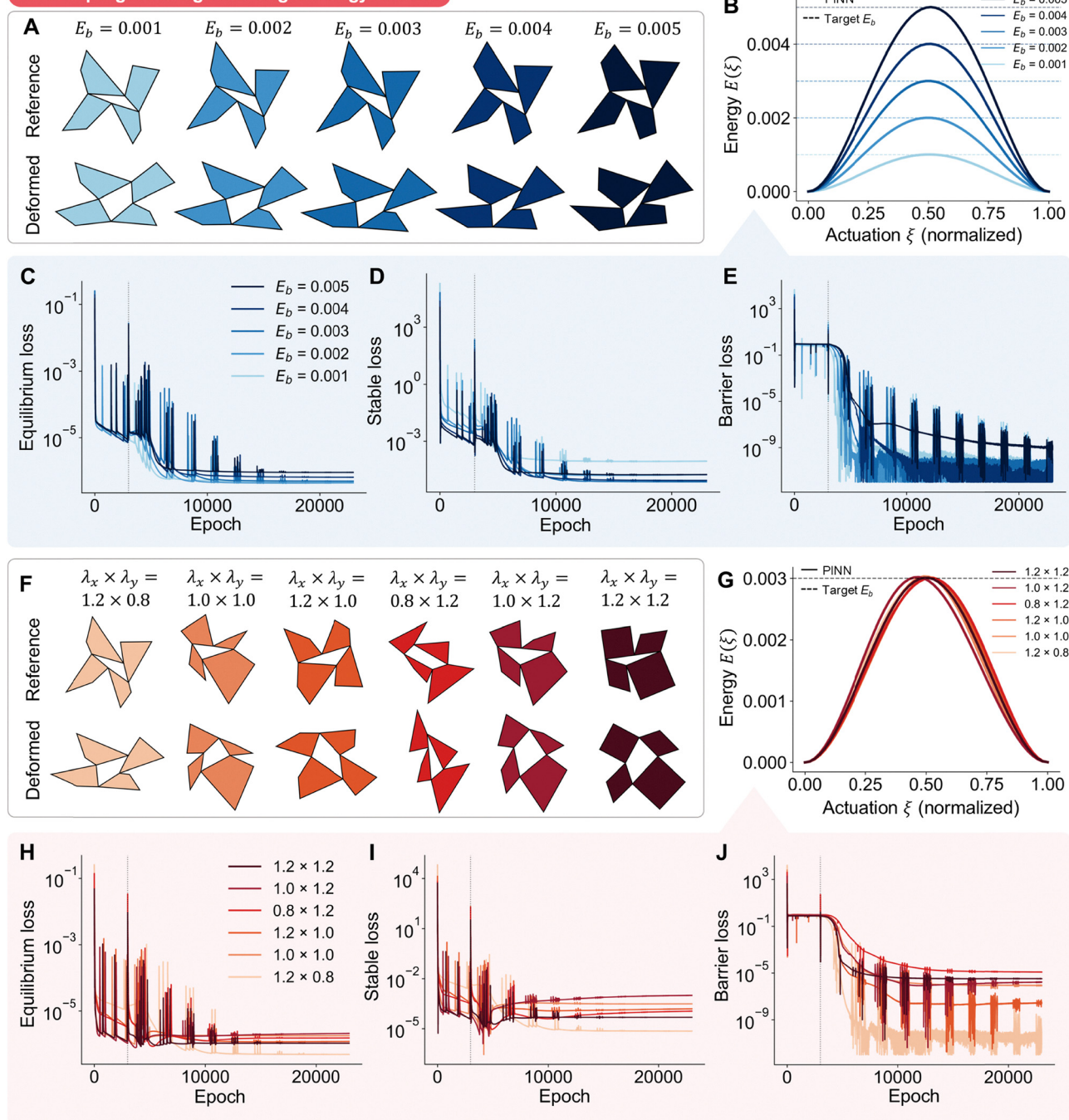


Fig. 5 Inverse programming of bistable kirigami energy landscapes from target energy barriers. (A)–(E) Inverse programming results under fixed macroscopic deformation with varying target energy barriers. (A) Reference and deformed configurations of five kirigami structures. (B) Programmed energy landscapes with predicted and target energy barrier values indicated. (C) Equilibrium loss histories for all five cases. (D) Stable-state loss histories for all five cases. (E) Barrier loss histories for all five cases. (F–J) Inverse programming results under fixed target energy barrier with varying macroscopic deformations. (F) Reference and deformed configurations of six kirigami structures. (G) Programmed energy landscapes with predicted and target energy barrier values indicated. (H) Equilibrium loss histories for all six cases. (I) Stable-state loss histories for all six cases. (J) Barrier loss histories for all six cases.

in Fig. 5F. The programmed energy landscapes (Fig. 5G) again reach the prescribed barrier values across all deformation conditions, despite substantial variation in macroscopic deformation. The associated loss histories (Fig. 5H–J) demonstrate stable convergence behavior across all cases.

Across all ten cases, the relative error in the energy barrier remains below 1.1% with a mean of 0.4% (SI, Section S5), while the stable-state energies are driven to the order of 10^{-9} , effectively reaching machine precision. Compared with the full-curve inverse design in Section 3.2, the barrier accuracy is

slightly improved (mean error 0.4% vs. 0.7%), as the barrier is directly targeted in the loss function rather than emerging indirectly from full-curve matching.

The reduced specification also increases the design freedom of the inverse problem, leading to geometries that differ more substantially from the original structures in Section 3.1 than those obtained from the full-curve inverse design. This behavior is consistent with the one-to-many nature of the inverse problem, where fewer constraints enlarge the space of admissible solutions. Additional multi-start analyses for both inverse formulations are provided in SI Section S8, further illustrating the non-uniqueness of the inverse solutions and the diversity of feasible geometries that satisfy the same energy target.

Finally, we compare the PINN framework with conventional constrained optimization based on sequential quadratic programming (SQP) with hard inequality constraints. For several cases, the conventional approach fails to reach the target energy barrier, whereas the PINN framework successfully satisfies the specification in all ten cases. Detailed comparisons are provided in SI, Section S6. The obtained design parameters are listed in SI, Section S5.

3.4. Finite element validation

To validate the inverse-designed structures beyond the reduced-order spring model, finite element analyses were performed on the five structures obtained from the inverse programming results in Section 3.3 with fixed macroscopic deformation $(\lambda_x, \lambda_y) = (1.2, 0.8)$ and energy barriers ranging from $E_b = 0.001$ to 0.005 . As described in Section 2.3, the simulations employ a continuum representation with finite-width line

hinges, hyperelastic material properties, and quasi-static implicit dynamic analysis to capture the snap-through instability.

Fig. 6A shows the deformed configurations of the five structures at the second stable state obtained from the FEA. The resulting geometries closely match those predicted by the PINN framework, confirming that the inverse-designed structures reproduce the intended bistable behavior in the continuum representation. Fig. 6B shows the barrier-state configurations for the lowest and highest barrier designs ($E_b = 0.001$ and $E_b = 0.005$), visualized using the von Mises stress field. In both cases, stress is strongly localized at the four hinge regions, with the higher-barrier structure exhibiting significantly larger stress concentrations, consistent with the larger elastic deformation required to overcome the barrier.

Fig. 6C presents the energy landscapes computed from the FEA. The monotonic ordering of energy barriers predicted by the PINN framework is fully preserved in the continuum simulations, and structures designed with higher target barriers consistently exhibit higher barriers in the FEA results. One difference between the spring model and the continuum model is that the second stable state exhibits a small residual energy in the FEA results. This arises from the finite-width hinges used in the continuum model, which undergo deformation even in the nominally stable configuration. Additional energy-decomposition analysis (SI, Section S7) reveals that the strain energy is primarily localized in the hinge regions and remains nonzero at the second stable state. Furthermore, the relative contribution of hinge energy increases as the programmed energy barrier decreases, resulting in a progressive reduction of the effective energy well depth for low-barrier designs. A parametric study of hinge thickness (SI, Section S7) shows that increasing hinge thickness

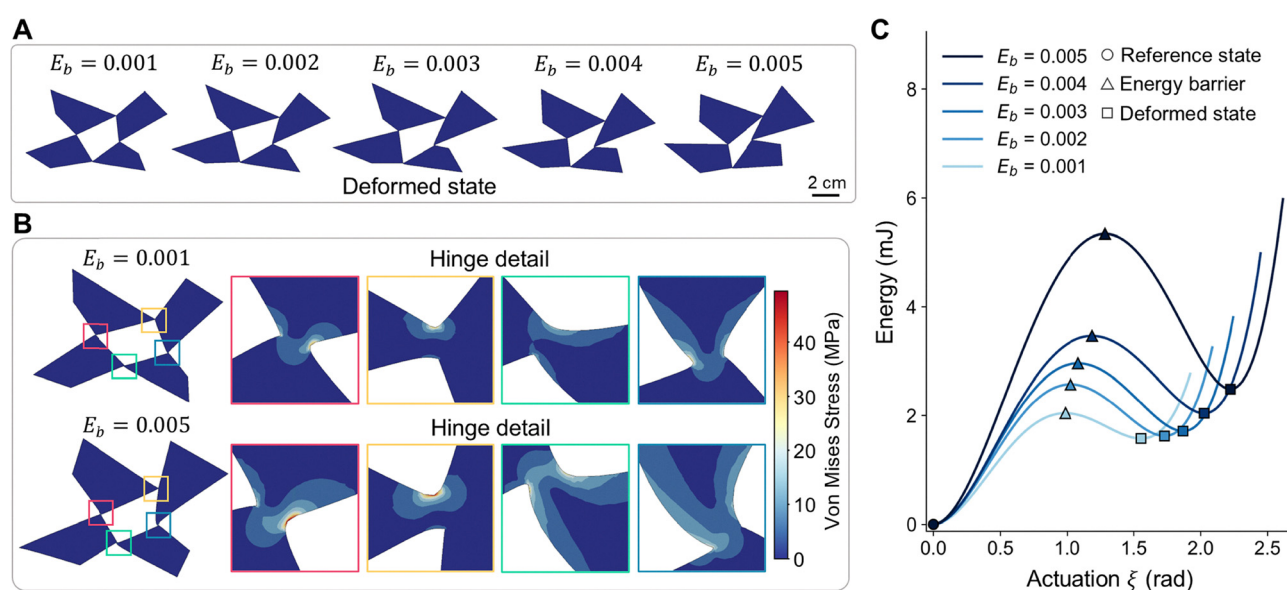


Fig. 6 Finite element analysis of inverse-programmed kirigami structures. (A) Deformed configurations of five kirigami structures at the second stable state, with target energy barriers ranging from $E_b = 0.001$ to 0.005 . (B) Von Mises stress distributions for the structures with the lowest ($E_b = 0.001$) and the highest ($E_b = 0.005$) energy barriers, showing the hinge locations and magnified views of the four hinge regions: top (yellow), bottom (green), left (red), and right (blue). (C) Energy landscapes obtained from FEA for all five structures along the actuation coordinate ξ , with markers indicating the reference state (circle), energy barrier (triangle), and deformed state (square).

gradually reduces the difference between the barrier energy and the residual energy and eventually leads to a transition from bistable to monostable behavior.

3.5. Experimental demonstration

To validate whether the bistable kirigami structures designed by the PINN-based inverse programming framework reproduce the intended hierarchy of energy barriers in physical specimens, five kirigami samples were fabricated by 3D printing as 2×2 arrays of unit cells with target barrier values ranging from $E_b = 0.001$ to 0.005 at intervals of 0.001 , while fixing the macroscopic deformation to $(\lambda_x, \lambda_y) = (1.2, 0.8)$. The fabricated geometries are shown in Fig. 7A. Manual actuation tests confirmed that the specimens designed with $E_b = 0.002$ to 0.005 stably maintained both the reference and deformed states, demonstrating clear bistable behavior. In contrast, the specimen designed with $E_b = 0.001$ returned spontaneously to the deformed state even after being manually transformed to the reference state, indicating monostable behavior. The corresponding demonstrations are provided in Video S1.

This discrepancy is attributed to the difference in hinge implementation between the design model and the fabricated specimens. In the PINN-based inverse design, the hinges were assumed to be point connections, whereas the printed specimens employed hinges with finite width, which introduce additional bending stiffness. As a result, when the structure is transformed from the deformed state to the reference state, prestress accumulates in the hinges. This interpretation is consistent with the FEA energy-decomposition analysis presented in SI Section S7, which shows that the relative contribution of hinge energy increases as the programmed energy barrier decreases. Consequently, low-barrier structures exhibit a reduced effective energy well depth and become increasingly sensitive to finite-hinge effects. In the physical TPU specimens, additional effects such as finite specimen thickness, out-of-plane deformation, and hinge torsion further amplify this trend. For the $E_b = 0.001$ specimen, whose designed energy barrier is relatively low, these effects have a proportionally larger influence on the overall energy landscape, contributing to the experimentally observed monostable behavior.

The force–displacement curves obtained from uniaxial tensile tests are shown in Fig. 7B. For the bistable specimens ($E_b = 0.002$ to 0.005), the measured force first increased to positive values during the initial stage of displacement-controlled loading, reflecting resistance to leaving the reference state. As the structure crossed the energy barrier, it spontaneously transitioned toward the deformed state and pushed back against the testing machine, leading to a negative force response. In addition, as the prescribed energy barrier increased, both the maximum positive force required for transition and the magnitude of the minimum negative force increased systematically. For the $E_b = 0.001$ specimen, a residual force was already present at the beginning of the test because the structure tended to recover from the reference state toward the deformed state. Nevertheless, during tensile loading, it still exhibited a similar sign change in force. The tensile tests are provided in Video S2.

The force–displacement curves were then integrated with respect to displacement to obtain the energy–displacement relations (Fig. 7C). Because prestress and stored elastic energy may remain in the reference state due to hinge deformation, the energy of the deformed state was set to zero, and the displacement was defined as the deformation length from the deformed state to the reference state. With this normalization, a clear energy barrier was identified across the full range of $E_b = 0.001$ to 0.005 , and the experimentally extracted barrier increased monotonically with the prescribed target barrier. To quantify the discrepancy between the PINN-programmed barriers and the FEA/experimental barriers, we quantified the preservation of the programmed barrier hierarchy after accounting for differences in energy scale, and bounded the residual discrepancy in SI, Section S7. This analysis shows that the programmed barrier hierarchy is preserved in both FEA and experiments (Pearson $r \geq 0.93$), with mean residual errors of 21.0% for FEA and 16.3% for experiment after scale alignment, and with the largest residual appearing at $E_b = 0.001$, where the energy well associated with the reference state was relatively shallow and only weakly separated from the barrier, consistent with the monostable behavior observed experimentally. To connect the programmed energy landscape to directly measurable mechanical properties, we further extracted the trigger-force metric and snap-through displacement from the measured force–displacement curves (SI, Section S4). The trigger-force metric shows a strong positive correlation with the experimentally measured energy barrier, confirming that the programmed energy landscape is translated into a measurable snap-through threshold. Beyond the scalar barrier, a point-by-point and shape comparison of the full normalized energy landscapes shows that the FEA profiles closely follow the PINN-programmed shape once the endpoint residual is removed. Without this correction, the FEA and experimental profiles deviate at the stable-state endpoints, reflecting the finite-hinge residual energy absent in the idealized point-hinge model; an energy decomposition into hinge and panel contributions confirms that this residual originates predominantly from the finite-width hinges (SI, Section S7B). In addition, the experimental profiles exhibit earlier peak positions than the PINN and FEA profiles. This peak-position shift is interpreted as arising from three-dimensional deformation effects in the finite-thickness physical specimens, especially out-of-plane deformation, which are not fully captured by the two-dimensional PINN/FEA models (SI, Section S7D). The programmed barrier hierarchy is nonetheless preserved in both FEA and experiment (SI, Section S7D).

We next investigate sequential deformation behavior enabled by the programmed energy barriers. A single specimen was fabricated as a 4×2 array, where three structures were sequentially arranged from top to bottom with $E_b = 0.003$, 0.004 , and 0.005 , while fixing the macroscopic deformation to $(\lambda_x, \lambda_y) = (1.2, 0.8)$. Uniaxial tensile tests were performed after zeroing the initial load to account for self-weight.

Fig. 7D shows the measured force–displacement curves. Overall, the force increased with displacement, with noticeable variations during structural transformation. The structure with

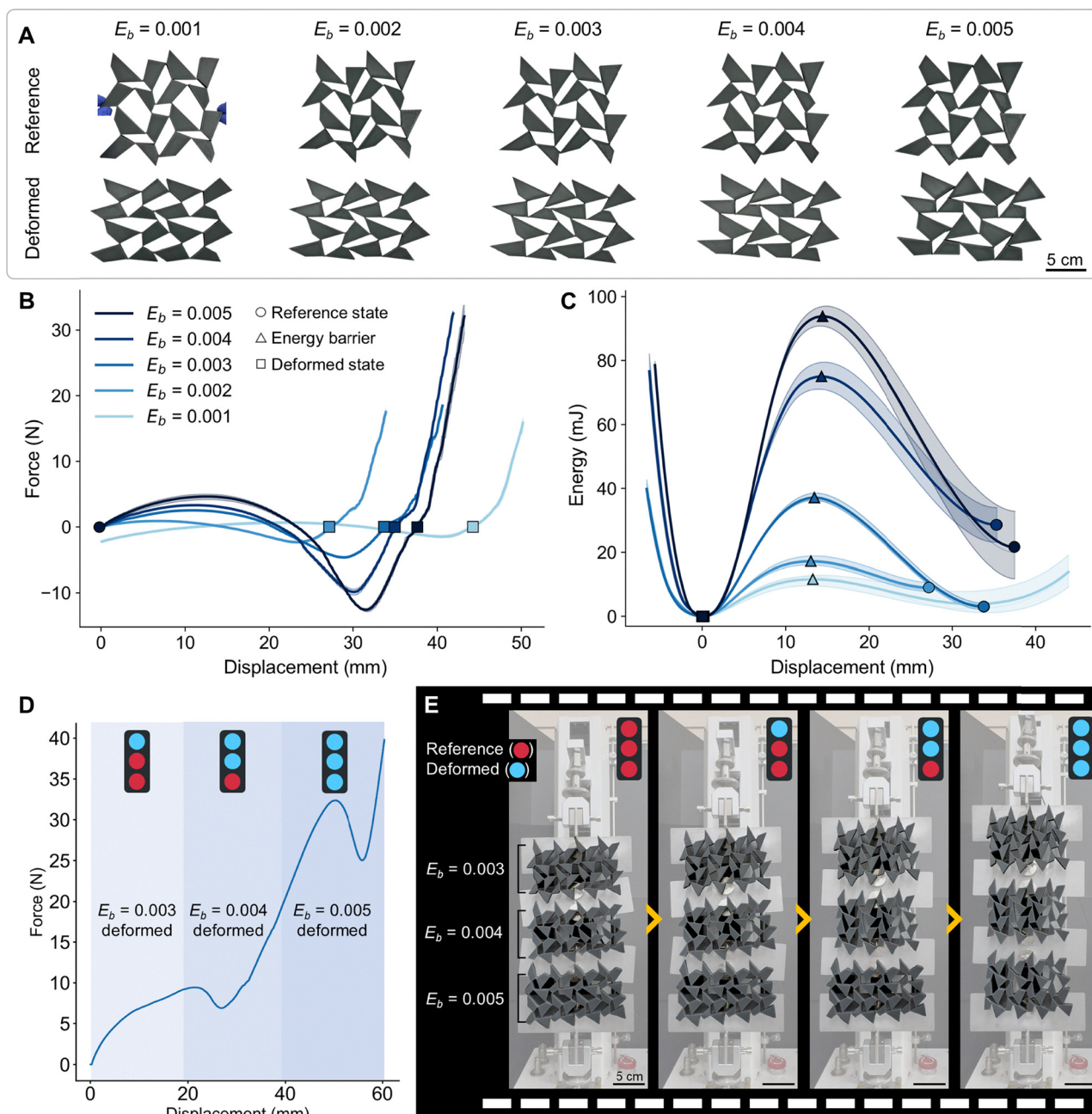


Fig. 7 Experimental validation and sequential actuation demonstration. (A) Photographs of 3D-printed kirigami specimens (2 × 2 unit cell arrays) in the reference and deformed states for target energy barriers $E_b = 0.001$ to 0.005 . (B) Measured force–displacement curves for all five specimens, showing the mean response ($n = 3$, where n denotes the number of experiments), with markers indicating the reference state (circle) and deformed state (square). (C) Corresponding energy–displacement curves, with markers indicating the reference state (circle), energy barrier (triangle), and deformed state (square); shaded bands indicate variation ($n = 3$). (D) Force–displacement curve of the three-unit assembly ($E_b = 0.003$, 0.004 , and 0.005) red and blue indicators denote the reference and deformed states, respectively, for the three units (top to bottom: $E_b = 0.003$, 0.004 , and 0.005). (E) Sequential actuation of the three-unit assembly, with units transforming in order of increasing energy barrier. Red and blue indicators denote the reference and deformed states, respectively (top to bottom: $E_b = 0.003$, 0.004 , and 0.005).

$E_b = 0.003$ exhibited a relatively gradual increase in force, whereas the structures with $E_b = 0.004$ and 0.005 showed a reduction in force during deformation. However, no transition to negative force was observed.

Nevertheless, as shown in Fig. 7E, the structures deformed sequentially according to the programmed energy barrier levels

(Video S3). This result demonstrates that the relative ordering of the designed energy barriers is preserved in arrayed configurations, confirming the feasibility of sequential actuation using multiple kirigami units.

This response is associated with the effects of structural scaling and hinge mechanics. In the present design, the overall

structure was miniaturized to accommodate multiple unit cells within a single specimen, while the hinge width was maintained, resulting in an increased relative hinge stiffness. Consequently, the characteristic snap-through behavior expected from the ideal point-hinge model was partially suppressed, leading to a reduced force drop during transition. To realize monolithic kirigami structures with clearer bistable responses, it will be necessary to more faithfully replicate the hinge behavior assumed in the design model, which may be achieved through improved printing resolution and refined hinge design.

4. Conclusions

This work presents a unified, data-free physics-informed neural network framework for predicting and programming bistable kirigami energy landscapes. By embedding equilibrium conditions, energy formulations, and geometric compatibility directly into the learning objective, the framework enables accurate forward prediction as well as inverse design under both fully specified and minimally constrained settings. The proposed approach achieves high accuracy across all tasks, with forward prediction errors below 0.1%, full-curve inverse design errors below 5%, and minimal-specification inverse programming achieving a mean barrier error of 0.4%.

Beyond accuracy, the framework introduces a new design paradigm in which energy barriers can be directly programmed with reduced input requirements. The minimal-specification formulation expands the design space and enables multiple geometrically distinct solutions for a given energy target, providing flexibility for practical implementation. Multi-fidelity validation through finite element analysis and experiments confirms that the relative ordering of energy barriers is robustly preserved, enabling reliable sequential actuation behavior.

These results establish a physics-informed approach for designing programmable bistable metamaterials based on energy landscape control. The ability to independently tailor energy barriers while maintaining a common deformation mode opens opportunities in deployable structures, soft robotics, and impact mitigation systems. Future work will focus on extending the framework to multi-cell architectures with enriched physical models that account for additional deformation modes, material-dependent limits, and environmental effects, enabling more realistic system-level mechanical design.

Author contributions

Sukheon Kang: conceptualization, data curation, formal analysis, investigation, methodology, software, validation, visualization, writing – original draft, writing – review & editing. Sukkyung Kang: conceptualization, formal analysis, investigation, methodology, validation, visualization, writing – original draft, writing – review & editing. Sanha Kim: funding acquisition, methodology, supervision, writing – review & editing. Seunghwa Ryu: conceptualization, funding acquisition, methodology,

project administration, resources, supervision, writing – review & editing.

Conflicts of interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

All code and data used in this study are openly available on Zenodo, accessible at: <https://doi.org/10.5281/zenodo.19350125>.

Supplementary information is available. See DOI: <https://doi.org/10.1039/d6mh00630b>.

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